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The Gap Between Technological Advancement and Gig Economy Legislation: A Case Study
Analysis

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ABSTRACT

Workers' compensation and rights within the gig economy are often a complete black box, with workers unable to understand how their pay was calculated, and riders not knowing the true fraction of their fare to the worker (Sathuluri 2026). In the new age of gig economy workers, our policy environment and legislation need to grow with our technological changes. We offer critiques on the current legislative environment with a case-study analysis of current gig economy legislation in Colorado, specifically after the passing of Senate Bill 24-075, Transportation Network Company (TNC) Transparency Act. The thesis employs a difference-in-differences framework on California and Colorado Uber driver data to analyze the impact of transparency policy on driver tips. The main conclusion of this analysis is that while there were no changes within the magnitude of tips or driver effort, there was a statistically significant increase in the proportion of zero-dollar tips after the treatment. This paper demonstrates the hardships of evaluating gig platform policy and the lack of accountability and auditing through legislation. Additionally, further examples and evidence of the limitations of the current policy environment are presented, and they highlight the importance of proper regulation on algorithmic management in the gig economy. The thesis demonstrates the clear lack of mechanisms for policy implementation and review within the ever-changing gig economy and highlights the work of independent researchers in holding gig platforms accountable for their actions on the growing labor force.

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Chapter 1 Introduction

In our current economy, the gig economy has flourished with technological advancement (Anderson et al 2021). The gig economy consists of all workers who rely on apps and websites to tell them when and where to work (Torpey 2016). From Uber drivers to handymen, many citizens are now turning to gig work as a source of income, if not their sole one, with surveys showing 16% of Americans earning money via online gig platforms (Anderson et al 2021). However, with gig work comes a multitude of issues. When managers become algorithms, there is no wage transparency, very limited work autonomy, and constant surveillance and control (Mate 2026). These apps are run with “fragmented communication channels” that make it even harder to fix these issues when workers cannot advocate for themselves to managers or leadership (Sendbird Industry Report 2024).

Since there is a clear disconnect between workers and the entities that control their wages and conditions, in place of negotiation, workers have turned to forming coalitions and advocating for legislation to regulate these corporations (Padin 2025). Recently, grassroots organizations, like the Workers Algorithm Observatory (WAO), have been created to work with technologists and workers to lobby for legislation that increases workers’ rights, safety, and wages. On January 1st, 2025, Colorado enacted legislation (Senate Bill 24-075), with the help of these organizations, requiring rideshare platforms to disclose fare and driver pay structures to riders and drivers (Colorado General Assembly 2024). Unfortunately, in this case, legislation is not the sole answer to the issue of workers’ rights and compensation in the gig economy.

Even though the bill included clear design mandates, companies like Uber were able to make major changes to their platforms to accommodate the mandated implementations (Cheshire

2025). While they argued these changes were necessary due to legislation, it was a clear exertion of power over their workers (Cheshire 2025). User interface (UI) design changes can potentially completely change driver conditions and compensation (NYC Consumer and Worker Protection 2026). In New York City, the change of the tipping option to after checkout cost delivery workers millions of dollars based on reports (NYC Consumer and Worker Protection 2026). There is clear evidence that the current policy environment for gig platforms is insufficient to create overarching design mandates that will benefit workers. The gap between what legislators ask of companies and the actual technology design changes is a significant issue in gig workers' rights and conditions (Padin 2025). More so, there are very few standards to uphold and evaluate legislation currently, and for proper regulations to be put in place, states need to "...create oversight units to license gig platforms, audit algorithmic practices, and ensure compliance" (Dumesnil 2026).

In this thesis, I will explore this gap in legislation through analytical research on the Colorado case study and present examples of similar legislative implementations regarding gig economy workers. This paper will present a difference-in-differences analysis of Colorado and California Uber drivers 12 weeks before and after the policy to determine whether there is an effect of the policy on drivers and how this policy impacts tipping. Our contributions offer insight into the real-world effects of transparency legislation in the gig economy and the impacts this legislation creates for workers within rideshare platforms. Additionally, the paper will examine further evidence of this legislation gap in gig work, such as cases from New York, Seattle, and California. Both parts of this thesis will illustrate the clear gap between legislation, implementation, and evaluation of gig work in our current environment and how they impact

workers. More so, it will be a critique of our current policy mechanisms and the lack of accountability on gig platforms for their implementation and changes in response to policy mandates.

Chapter 2 Literature Review

Gig Economy Overview

While the gig economy is relatively new, there has been emerging research and studies done on the field, the workforce, and the algorithms surrounding these platforms. Research done by the Congressional Research Service defines the gig economy as a “collection of markets that match providers to consumers on a gig (or job) basis in support of on-demand commerce” (Congressional Research Service 2016). While the data analysis in this thesis pertains to rideshare platforms, there are many other uses of the gig economy. An important distinction for gig workers surrounding legislation is that they are treated as independent contractors (Mate 2026). This prevents these workers from receiving benefits and protections from the companies they work under, such as healthcare, workers' compensation, etc. (Congressional Research Service 2016). The Congressional Research Service points out 3 distinct features of gig work that differ from traditional self-employment work. These features include platforms taking commission from the transactions that occur on the platform, platforms having the right to terminate or end worker contracts as they deem fit, and platformers controlling relationships between workers and their clients (Congressional Research Service 2016). These differences increase the control platforms have on gig workers, all happening concurrently with workers being viewed as independent contractors (Mate 2026).

Policy Regulations on Companies

Gig platforms are not the only types of companies that are regulated by the government, and most industries within the US and the global economy have been regulated in some way, shape, or form throughout history (Vietor 2008). Companies tend to have the freedom to make

decisions and changes that are necessary to meet mandates or regulations (Schaffer 1995). Economic policies and how they impact firms are extremely relevant when evaluating the gap between legislation and company implementation. This relationship is rooted deeply in economic theory and firm behavior to maximize profit (Oliver 1991). Firms and industries react and respond to regulations in different ways, mainly strategic adaptation and political influence, as seen in the paper, “Firm-level Responses to Government Regulation: Theoretical and Research Approaches”. Many parallels from the paper can be drawn from these firm behaviors to Uber’s actions after Colorado’s transparency policy. The paper states, “When government uses its authority to implement public policies through processes of legislation and regulation, the main response of firms is adaptation” (Schaffer 1995). The statement holds with the theory presented in “Strategic Responses to Institutional Processes” by Christine Oliver. She mentions multiple different responses to institutional processes and states, “...when anticipated legitimacy or economic gain is low, organizations will attempt to compromise on the requirements for conformity, avoid the conditions that make conformity necessary, defy the institutional requirements to which they are advised to conform, or manipulate the criteria or conditions of conformity” (Oliver 1991).

The manipulation is a key function of the current gig economy in response to regulation and aligns with the theory where companies take regulations, and they are “...designed and operated primarily for its benefit” (Stigler 1971). This is highlighted in this case study, where Uber conformed with transparency law, but also made important design changes to benefit their algorithm (Cheshire 2025). In addition to this approach, Schaffer’s paper also researches firms lobbying and using political influence to change policy in their favor (Schaffer 1995). Both

responses can be demonstrated in Colorado's example, with lawsuits against Colorado's government to change policies by Uber and UI/design changes to follow the strategic adaptation of transparency regulations. Uber's lawsuit claimed that Colorado had violated their "free speech and free association under the First Amendment of the U.S. Constitution", since they were being compelled to share fare information (Wiessner 2025). While this was not a successful lawsuit, it is consistent with the theories of the paper in utilizing influence and power in the political space.

Gig Worker Regulations

Regulations on the gig economy have come in many specific areas; however, few changes have been made on the federal level to increase regulations and protect workers' rights (US Department of Labor 2026). Recently, federal law has been changed to view workers under gig platforms as independent contractors, which prevents workers from certain rights under the Fair Labor Standards Act (US Department of Labor 2026). While these changes have been made in America, a paper done by researchers in Australia points out 5 key regulations/changes that need to be made in the gig economy (Dubal 2017). While these are in a different legislative landscape, it highlights the future of legislation within the gig economy and factors to consider when examining Colorado law SB 24-075. These 5 areas consist of: confirming/enforcing existing laws, clearly defining employment, creating a new category of independent worker, creating rights for workers instead of employees, and reconsidering the concept of employer (Dubal 2017). Within these 5 areas, the most applicable to this thesis is the confirmation and enforcement of existing laws. Currently, the mechanisms to make sure legislation is beneficial and truly being enforced are not sufficient for the current needs of workers due to a lack of transparency (Mate 2026). In the paper "Algorithmic Management and Workers' Rights in the

Gig Economy: Legal Challenges and Regulatory Responses in Digital Labour Markets” the author states, “The relative novelty of algorithmic governance, combined with the deliberate organisational strategies of platforms designed to avoid employer status, has led to inconsistent judicial outcomes and fragmented regulatory approaches” (Mate 2026). Whether there are gaps in data, misrepresentation of policies, or additional UI/algorithmic changes, there can be numerous factors that change how gig workers are being regulated between legislation and implementation.

Legislation Evaluation Research

As mentioned, there are many factors for laws to be ineffective or lead to unintended consequences in the gig economy. This furthers the argument that regulations must be evaluated and that legislation gaps form when frameworks are not in place for these evaluations. Previous research has been done on post-implementation reviews (PIRs) for regulations made by the government (Katz 2016). PIRs are used to “... allow policy makers to see if regulations are operating as expected and achieving the intended objectives, or alternatively if they are leading to unintended consequences or imposing disproportionately high costs relative to the benefits that they offer” (Katz 2016). While PIRs exist throughout different government structures in the world, in the United States, they operate as individual federal agencies conducting their own analysis and reporting (Sozan and Durudogan 2025). In the case of SB 24-075, Colorado’s General Assembly asks for data collection from TNCs 18 months after the policy went into place (Colorado General Assembly 2024). The issue with this system under the gig economy is the lack of transparency of algorithms and access to data, which would make it harder to conduct PIRs or any enforcement and evaluation within our legislative system (National Employment

Law Project 2025). In addition, regulations that are being implemented may be met; however, there must be overarching assessments to include the additional changes that can impact different parameters/metrics (Sathuluri 2026). In this thesis's case study, Colorado's regulation surrounds transparency; however, it is important to look at workers' compensation and tips as well to ensure the policy didn't negatively impact workers with the UI changes Uber made. This further highlights the importance of comprehensive legislation evaluation in the gig economy.

Chapter 3 Examples of Gig Economy Legislation and Responses

There have been multiple examples of gig economy legislation in New York, Seattle, and California. Each of these examples highlights companies finding loopholes to increase profit when faced with legislation intended to increase workers' rights and conditions. They also clearly demonstrate the lack of oversight within the policy-making process, since these actions and effects were all independently researched outside of the legislative process.

In New York City, there have been major legislative changes both with rideshare platforms and delivery worker platforms. In 2019, the New York City Taxi and Limousine Commission adopted legislation that guaranteed a pay standard of \$17.22 an hour (Center of New York City Affairs 2019). The Center of New York City Affairs reported that “96 percent of Uber, Lyft, Juno/Gett, and Via drivers (i.e. 76,800 drivers) will receive an annual average increase of nearly \$10,000 under the minimum pay standard, which equates to a 45 percent net pay increase” (Center of New York City Affairs 2019). The law's original intent was to help all drivers earn a minimum wage. However, Uber and Lyft pivoted to locking out drivers from the apps for periods of time (Robinson 2025). Since they were unable to use the app, it increased the utilization rate, which is the percentage of time drivers have passengers when working (Robinson 2025). The commission's law stated that if drivers had a lower than 53% utilization rate, rideshare platforms were required to increase their pay to drivers (Center of New York City Affairs 2019). The increase in utilization rate from lockouts helped companies avoid paying their drivers the mandated amount (Robinson 2025). This is a clear example of the ways rideshare platforms can game the system and legislation made to enforce adherence to minimum pay (Robinson 2025). While legislation could not have foreseen circumstances like lockout periods

to increase utilization rates, there was a gap between companies and legislation for these policies to be carried out properly.

Additionally in New York City, a similar situation occurred with delivery drivers. After legislation was passed by the New York Supreme Court that apps had to pay a minimum pay rate of \$17.96 per hour in 2023, Uber and DoorDash announced plans to remove the tipping prompt until after a delivery is completed and include a service fee (McCarthy 2023). This led to a dramatic difference in average tips for workers and extreme negative consequences for their pay. Based on information from the New York City Department of Consumer and Worker Protection, Uber Eats and DoorDash tips were at an average of \$0.76 after the change was put into place, while for other restaurants without these changes, tipping is around \$2.17 (NYC Department of Consumer and Worker Protection 2026). The decrease in tips is estimated to amount to \$554 million for workers (NYC Department of Consumer and Worker Protection 2026). A clear mandate on price minimum completely changed the user experience, leading to a decrease of thousands of dollars for the average delivery driver.

In Seattle, a very similar example played out for gig workers. In 2024, legislation was made to create a minimum wage mandate for gig workers (Healy 2024). Drivers reported seeing increases in wages at first, but after a few months, orders stopped coming in for drivers, prices skyrocketed for consumers, and workers' compensation had decreased significantly (Healy 2024). According to researchers at Carnegie Mellon University, "If the market for drivers is indeed subject to nearly free entry, minimum pay policies will struggle to raise drivers' earnings without imposing some form of entry barrier" (Heinz College 2026). In addition to the influx of workers, the researchers also found decreases in tips, fewer tasks each day, and longer driving

distances (Heinz College 2026). This shows that there were additional algorithmic changes in conjunction with legislative mandates. Ultimately, the changes in legislation caused the opposite of the intended consequences for workers.

A final example is Proposition 22 in California, which was created by rideshare platforms to view gig workers as independent contractors. While this stripped away many of their rights as “full-time employees”, it obligated companies to provide them with 120% of the minimum wage and subsidies for healthcare for those who work around 15-25 hours per week (Sainato 2021). Research done by UC Berkley’s Labor Center found that instead of the \$15.60 an hour wage that would be required under the law, the estimated amount was the equivalent of \$5.64 an hour (Jacobs and Reich 2019). Research found companies finding numerous loopholes to stay within regulations but not pay drivers more, the most prevalent of them being only counting engaged time when drivers are actively driving passengers (Jacobs and Reich 2019). With this change, companies were able to severely manipulate the amount of time that was being compensated and therefore recognized under the law (Jacobs and Reich 2019). Additionally, the same tactic was used to prevent individuals from reaching the 15-hour minimum by only providing them with rides a percentage of their time online and making it harder for them to work their intended hours (Jacobs and Reich 2020). Even when workers were waiting on the app for jobs/orders, their time waiting would not be recognized as “engaged time” so that Uber and other companies would not be mandated to view them as employees under their applications (Jacobs and Reich 2020). This continues to show the clear trends of finding workarounds within legislative systems and implementing around mandated parameters for their own gain and bottom line.

Chapter 4 Colorado Bill Overview

The Senate Bill 24-075, known as the Transportation Network Company (TNC) Transparency Act, was passed in Colorado on June 5th, 2024, and went into place on February 1st, 2025. There were many different aspects of the bill related to TNCs, mainly deactivation standards for drivers and transparency for both consumers and drivers. Specifically, the main areas under transparency for driver fares and tips come from Section 11. This states that a TNC must disclose the following to a driver: the total amount of money that the TNC will pay the driver, the aggregate estimated mileage that the driver will drive, the amount of tip if already selected, the total amount the customer paid before tip, and the total amount the driver will receive before tip (Colorado General Assembly 2024). To a consumer, the bill states that TNC's must disclose the total amount of money the consumer paid and the total amount that the driver will receive without a tip.

With these requirements, certain design elements were explicitly stated, including font size, single-screen visibility (all information in one place), a cohesive data interface for the driver to see all aspects, and downloadable data tracking (Colorado General Assembly 2024). While these aspects are helpful and prevent companies from hiding information, they do not go into depth on how/when information will be presented to consumers or if other changes can be made in conjunction with the policy mandates. This allowed TNCs to make their own modifications and changes that ultimately better their profit margins and algorithms, even with the proposed changes (Cheshire 2025).

In the implementation guide of the bill, there are further requirements and deadlines that need to be met. The most notable thing from this guide is the mandate of semiannual disclosures. The bill specifically states, “On a semiannual basis beginning August 1, 2026, a TNC is required to disclose various information to the DLSS, including detailed information on drivers, driver deactivations and suspensions, and transportation tasks for which the TNC dispatches a driver” (Colorado General Assembly 2024). This puts into place a data collection process for the government; however, a key point is that it does not necessarily indicate what information will be collected, and TNCs will be able to provide what information they deem fit, which can be skewed or manipulated for their agenda

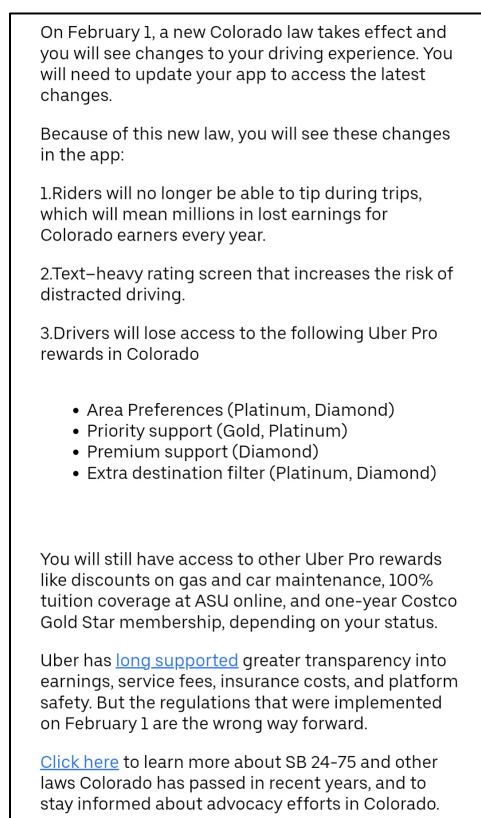


Figure 1. Uber’s message to drivers after Colorado SB 24-075

There were many prominent changes that were made to the user interface after the law went into place. Figure 1 highlights the statement made to riders describing the new changes Uber made in response to the law. The largest and most impactful design change was turning off the ability to tip during a ride for Colorado users (Cheshire 2025). Uber blamed the policy, stating, “We made these changes to comply with the law.... Because the total fare is not final before the trip is over, we could not allow riders to tip until after the trip is over. ...During the legislative process, we warned that this law would have unintended consequences and this is one of them” (Cheshire 2025). They also removed key reward systems like “area preferences, priority support, premium support, and extra destination filters” (Cheshire 2025). These changes were not mandated by the legislation, and these features make it much more difficult to truly analyze the results of this bill, specifically regarding how it affects driver tips, as analyzed in the next section.

Chapter 5 Methodology

Analysis Overview

While Uber does not provide driver data to the public, researchers at FairFare have been able to collect data through connecting with unions and drivers across the country. This paper uses Uber driver data from Colorado and California to conduct a difference-in-differences analysis. To start, the setup of this analysis starts with 2 groups, treatment and control, and 2 periods, before and after treatment. Difference-in-Differences is an analysis to view “how much more the treated group changed than the untreated group when going from before to after” (Huntington-Klein 2021). In this case, Colorado is considered the treated group, and California is considered the untreated group, since SB 24-075 was only passed for Colorado. The purpose of this study is to examine the change in income tips in Colorado compared to California due to the change in policy. In the most basic DiD design, *The Effect* by Nick Huntington-Klein mentions the assumptions that are most needed: parallel trends and untreated groups (Huntington-Klein 2021). The untreated group satisfies the parallel trends condition. This condition states that the trends between the treated and untreated groups would stay the same if there were no treatment. A key assumption being made is that Colorado would have similar trends in income tips if left untreated. In our implementation of the DiD study, we utilize the treatment being put into place on February 1st, 2025, and we use 12-week time spans for the before and after periods.

Code Description

To give a brief overview of the data, the 'argyle_activities_all_fields' database includes all information about driver data, the rides, corresponding income, and tips that are transacted.

This was collected from drivers through FairFare and contained driver data for 58 drivers within Colorado and 35 drivers within California. Below is the breakdown of key variables that were used within the data and definitions of the fields.

Attribute	Description
id	Unique ID of the gig assignment
account	Unique ID of the account associated with the gig assignment.
start_location_lng	Longitude (ISO 6709) where the gig assignment started
start_location_lat	Latitude (ISO 6709) where the gig assignment started
end_location_lng	Longitude (ISO 6709) where the gig assignment ended
end_location_lat	Latitude (ISO 6709) where the gig assignment ended
start_location_formatted_address	Full address of where the gig assignment started.
end_location_formatted_address	Full address of where the gig assignment ended.
income_pay	Earnings related to the assignment received by the gig employee
income_tips	Tips received by the gig employee
income_customer_price	Amount the customer paid for the order, not including fees and tips. Only available if the gig platform provides this value.
income_total_charge	Total amount charged to the customer
income_fees	Fees charged to the gig employee by the gig platform
income_total	total earnings received by the gig employee (pay + tips + bonus + other)
start_datetime	Timestamp (ISO 8601) when the gig assignment started
end_datetime	Timestamp (ISO 8601) when the gig assignment ended

Table 1. Argyle Activities Data Breakdown

For data cleaning, the data was filtered to only include 12 weeks before (November 9, 2024) and after (April 26, 2025) the policy date of February 1st, 2025. Additionally, cancelled

rides were filtered out, and only inner state rides within Colorado and California were included. New columns were created to find the income tips as a percentage of the customer price and the tip as a percentage of the driver's pay. Additionally, the gap between trips, essentially the driver's wait time between rides, was measured as a function of start and end times for their trips. To help with the creation of treatment periods, weeks were assigned for each data point from -12 to 12. The overall trip time and distance were also calculated. Lastly, the main data frame was separated into Colorado and California data sets.

Appropriately with Difference-in-Differences testing, the control and treatment groups were determined by state and time period. Driver transactions that were in the state of California, regardless of date, are in the control group. In addition to these data points, Colorado transactions that were prior to the policy date were considered in the control group. Only Colorado transactions that were after the policy date were in the treatment group. A new data set was created by grouping all transactions by driver account and week to aggregate a driver's average income tips during a given week. The interaction variable was created by looking at the interaction between time and state, creating values of 0 or 1 based on the treatment group. The control group resulted in a 0-interaction value, while treatment was equal to 1. This process was then repeated to create another data frame with aggregate income tips as a percentage of customer price.

Once the data was isolated into treatment groups, an Ordinary Least Squares Regression was performed based on the time and state, and its effect on the weekly mean of income tips per driver. The following equation shows the regression function being used.

$$incometips = A_0 + A_1 * Time_i + A_2 * State_j + A_3 * Time_i * State_j$$

This regression was also run on the second dependent variable of income tips as a percentage of customer price. The variable “tips_as_percent_of_customer_price” was used to control for fare prices increasing with additional fees or inflation, and it was used to normalize tips in relation to the length and price of the drive. In addition to the regressions that were run, different visualizations were created to view trends of both California and Colorado weekly tips.

The Event Study Specification was a dynamic difference-in-differences test conducted to create interactions for each time period in the analysis. This was done by regressing on the "tip rate", or income tips as a percentage of the customer price.

$$tip_percent = \alpha + \sum (\delta_t \times D_t \times \text{Colorado}) + \text{account_FE} + \text{week_FE} + \varepsilon$$

This type of analysis enables us to more granularly examine if an effect is happening, how soon the effect appears, how the effect size changes in relation to time, and the lasting nature of the effect over time. However, an important note is that this method would be more beneficial with a larger sample size in the future.

While regressions view relationships between treatment factors and the target variable of tips, it was necessary to measure the efforts of drivers. This is to ensure that even if driver tips are unaffected by the treatment, varied factors are not increasing or changing, which would impact drivers. In the examples of gig economy legislation, drivers were having longer wait times between rides and working harder for similar or less compensation. The main values to quantify effort within the data were trip time, distance driven, time between each trip, and

number of trips. To measure a change in these efforts, a t-test was run for each variable in the Colorado transactions grouped by account and week.

Lastly, raw income_tip amount changes before and after the policy in Colorado were calculated and compared. This was then converted into a percentage of non-zero tips transactions within all trips before and after the policy. A z-test was also conducted to see if the proportion of zero-dollar tips had a statistically significant difference between the treated and untreated groups. Additional visualizations were created to view changes in average tips on a week-by-week basis, both for Colorado and California. All these methods helped analyze the full impacts of the policy within Colorado and isolate any specific trends/patterns within the data.

Chapter 6 Results

The OLS regression for income tips resulted in a negative coefficient for time (-0.91), and positive coefficients for both state (0.25) and the interaction variable (0.90). The negative coefficient for time was statistically significant within the OLS regression. This can be interpreted as a negative relationship between income tips and time. Since time only has two values, 0 and 1, this means that tips generally decreased after the policy date on February 1st. As for the positive coefficients, neither variable was statistically significant, meaning transactions occurring in Colorado were not different from those in California, and there is no evidence that the treatment impacted income tips.

Differing from this regression, the OLS regression run on tips as a percentage of customer price showed no statistically significant correlations between the target variable and time, state, or the interaction variable. This difference between the percent tip and income tips shows that changes in tips would be attributed to longer distances or fares rather than changes in consumer behavior or driver effort.

This is consistent with the findings of the driver effort t-tests conducted. For every test, there was no statistically significant change between the before and after averages of trip time, trip distance, time between trips, or number of trips. This serves as evidence that there were no significant changes in driver effort after the policy went into place, at least within the 12-week post-treatment period.

With the dynamic difference-in-differences, the lead-lag visualization shown in Figure 2 concludes that the effect of the policy does not occur until 5 weeks after the policy went into place. After this point, it comes back to around 0, meaning the impact of the policy did not take effect immediately and only for a short amount of time. However, an increase in sample size would strengthen the statistical power of this visualization and test. The constraints of collecting driver-side data prevented researchers from dramatically increasing the sample size.

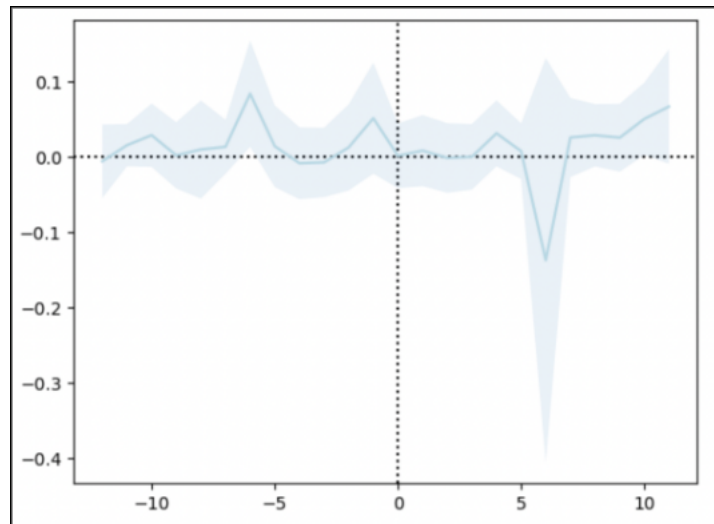


Figure 2. Lead-Lag Visualization from Dynamic DiD

As mentioned, raw variables before and after treatment averages were calculated and are shown below. There are slight changes within the majority of these values; however, considering the data sample, this does not signify any important changes within these values due to the treatment.

Variables	Average Before Treatment	Average After Treatment
income_tips	2.731841	2.609124
income_pay	17.22199	16.809543
income_total	20.870653	20.282031
income_fees	5.50283	7.115735
income_customer_price	29.30565	29.74169
income_total_charge	32.037491	32.350815
tip_as_percent_of_customer_price	0.093204	0.088684
tip_as_percent_of_driver_pay	0.157787	0.161148

Table 2. Averages Before and After Treatment

Visualizations to further examine trends throughout the 24-week period before and after the policy can be seen in the following graphs. The average tips are summed per week, aggregated over all drivers, and separated for California and Colorado drivers. The graph shows abnormally high averages for Colorado in week -4. California also decreases the average tips after the policy date compared to Colorado's average tips.

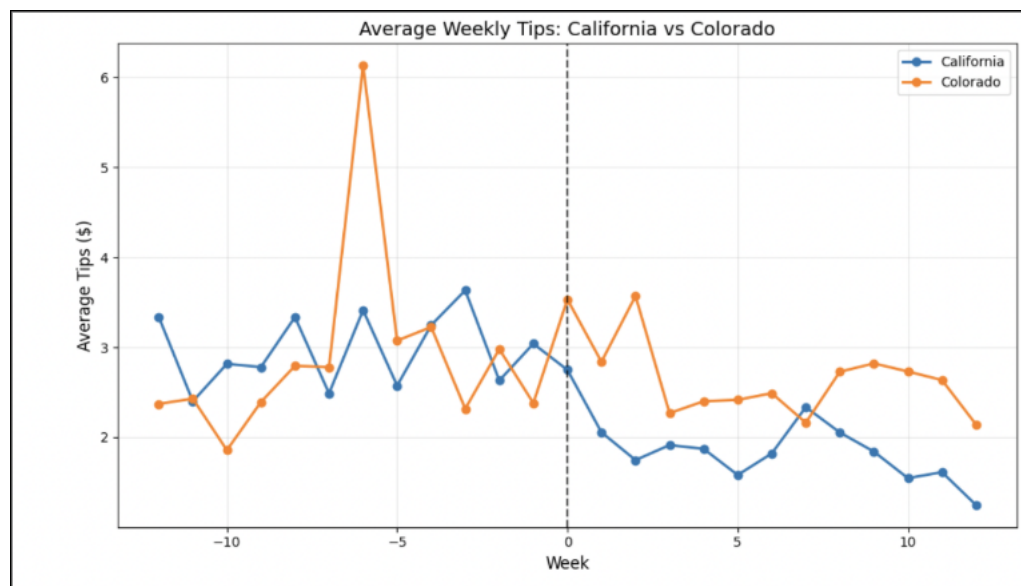


Figure 3. California vs. Colorado Average Weekly Tips

Weekly averages for each Colorado driver per week were calculated and compared between periods. Additionally, the percentage of rides that had positive, non-zero tips was 41% before the policy and 39% after the policy, showing a decrease in tipping behavior within Colorado. In conjunction, a statistically significant difference in the proportion of zero-dollar tips was found through the z-test. The OLS analysis found a non-significant relationship between tips and state and time, which suggests that there is no relationship between the treatment factors and tips. However, the difference in zero-dollar tip proportions is an important result, especially with the UI change of making it impossible to tip until after the ride is over, which may have prevented users from leaving a tip at all.

Overall, the main results of this analysis show that while the magnitude of income tips did not see a statistically significant shift, the frequency of tipping did. The increase in the proportion of zero-dollar tips could mean the policy reduced the likelihood of a passenger tipping at all. While driver behavior remained consistent and there were no statistically significant findings in the OLS between the treatment and income tips, the non-significant results further underscore the need for larger datasets to capture these nuanced changes with greater precision.

As a note, extended analysis from collaborators Samantha Dalal and Mark Tremblay was completed using both the current data set and consumer-side data. With the larger data set and utilizing Lyft data as well, they were able to find an average monthly loss in tips for Uber specifically, with a decrease of \$400,00 per month in tip revenue following the policy change. They concluded a decrease in the likelihood of tips occurring, similar to our conclusions, and were able to quantify the loss in tip revenue for Colorado drivers.

Chapter 7 Discussion & Further Research

The case study analysis conducted within this paper is much different than the previous examples cited within gig economy legislation. While Uber changed critical design features in response to the transparency law, significant negative impacts of this were not found within the analysis regarding tipping magnitude or driver effort. However, negative impacts were found in the increased proportion of zero-dollar tips. The analysis also demonstrates the difficulties in isolating the evaluation of policy change. There is no documentation of the actual changes Uber made to its platform and when. Driver-side data needed to be collected by third parties, making it increasingly difficult to have robust sample sizes, and without the work of independent researchers, the government mechanisms intended to evaluate this policy would not go into effect until August of this year. Not only are current regulations not being properly evaluated, but there is also very little transparency for independent parties to audit algorithmic management. Tech Policy states that algorithmic governance without regulation is “a tool for eroding democratic accountability” (Sathuluri 2026). Gig platforms can make changes as they see fit to their design and algorithms, which can potentially leave workers with no source of income, with little to no accountability (Sathuluri 2026).

The lack of transparency between gig platforms and the public is highlighted by the extreme difficulty in accessing driver-side data. Data collected for this study was done purely through the Workers Algorithm Observatory and their partnership with the Colorado Independent Drivers United union, where they get in touch with workers and utilize their accounts to gather data. Gig platforms make it virtually impossible to gather the data of their

workers and consequently harder to conduct studies on changes. This negatively impacts the legislative process and evaluation of regulations on gig platforms (National Employment Law Project 2025). The lack of transparency and accessibility surrounding driver data further demonstrates the inability to have full governance of the current work environment and compensation of workers.

Additionally, with PIRs and SB 24-075's mandate for driver and deactivation data, there is a clear gap between the current policy mechanisms and gig platform accountability. While data collection and reviews may work with evaluating other regulations, with the ever-changing space of platform work and environments, it is not enough to accurately uphold and evaluate legislation (National Employment Law Project 2025). Colorado's TNC data will not be shared until a year and a half after the policy was enacted. In that time, these platforms can completely change, making it impossible to determine the effects of the legislation. It may be unfeasible to isolate the changes to these apps during analysis, especially when apps update and change outside of legislation. In this case, the change in tipping screen after the ride is complete is related to transparency on the fare, and therefore, should be clearly documented and considered when evaluating laws. Technological platforms change too quickly for a legislative process to keep up, and according to Gary Marchant in *The Growing Gap Between Emerging Technologies and Legal-Ethical Oversight*, "...existing regulatory systems and ethical frameworks are inadequate to provide effective, meaningful and timely oversight of the current and future generations of emerging technologies" (Marchant 2011). While current labor regulations in other industries can be enforced through worker reports and Occupational Safety and Health Administration (OSHA) or US Department of Labor inspections, a gig worker has no physical

site to inspect, and no agency like OSHA equipped to handle the algorithmic managing practices. As of 2026, OSHA states that self-employed workers are not covered by their policies and protections. The combination of lags in regulation evaluation and the inability for in-person evaluation of algorithmic decisions creates unchecked control for corporations and a lack of accountability from the government.

Overall, this thesis is a clear representation of the current disconnect within the policy environment of the 21st century. The gig economy is ever-changing and becoming a large portion of the overall labor market (Anderson et al 2021). Companies will always prioritize their bottom line, and economic theory will come into play to regulate markets, but regulations still must be properly put into place and protect workers and their rights (Stigler 1971). The gig economy is currently being operated without proper oversight, and the examples provided in this thesis highlight this. With many citizens having gig work as their sole income, unfair deactivations, tipping changes, and algorithmic shifts all decrease transparency and fairness within a black box system (Mate 2026). While the current legislation environment classifies workers as independent contractors, corporations have full control over workers through algorithmic management, and there are “unaccountable power structures that operate outside democratic institutions” (Sathuluri 2026). Workers can only advocate for themselves through legislative change due to the nature of the gig economy; however, these regulatory bodies are not properly enacting and evaluating legislation, which can be seen through this case study. This analysis found a decrease in the likelihood of tipping, and it underscores the difficulties in isolating and detecting platform changes and the lag in which evaluation is being done of necessary regulations. Instead of our current system, which can only independently investigate and piece together the changes and

data of gig platforms, mechanisms must exist to examine data, algorithms, UI changes, and technology implications. As the gig economy continues to grow, platform legislation will become a prominent check on the companies that employ thousands of workers. This legislation will only be successful by closing the current divide between technology and the law.

Chapter 8 Conclusion

In conclusion, the findings of this paper demonstrate the clear disconnect between the intended effects of current policy legislation and platform decisions in the gig economy. Colorado's analysis shows the difficulty in finding the true impact of legislation on tipping when multiple factors are impacted, and user interfaces are drastically changed. However, even without the ability to isolate these changes, Colorado proves that simply mandating greater transparency in the gig economy is not the solution to better workers' rights or compensation. Legislation for the gig economy must be specific and methodological for design requirements that do not have loopholes for companies to jump through. This is key for a well-regulated and accountable gig economy.

As more workers join the gig economy, legislation must evolve to extend protections to a growing percentage of the workforce. Objective-based testing on gig platforms must also be easier to observe and conduct to understand the impact of legislation on workers and change mandates based on the conditions they undergo. The current environmental policy needs to be overarching in holding corporations accountable for the multitude of design and algorithmic changes that can drastically impact workers' compensation and conditions on the day to day. From lockouts to fewer tips, their ability to survive comes from algorithmic decisions that must be regulated. Future work in this area can pertain to the solutions towards this issue for legislators and analysis on other areas, and examples of gig workers being affected by policy and design implementations.

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